

Twenty-sixth day of class - 11/25

Fun fact: Let $E : y^2 + xy = x^3 - x^2 - 79x + 289$. Is $L''(E, 1) = 0$? We don't know. Magma verifies that $|L''(E, 1)| < 10^{-100}$.

This elliptic curve has rank 4 and the Birch and Swinnerton-Dyer conjecture predicts that

$$L(E, 1) = L'(E, 1) = L''(E, 1) = L'''(E, 1) = 0$$

while $L^{(4)}(E, 1) \neq 0$.

In general, the problem of finding an algorithm to compute the smallest k so that $L^{(k)}(E, 1) \neq 0$ is open. There would even be significant consequence if someone were to prove that $L''(E, 1) = 0$ for this particular curve E .

Note: Today is the last day of content. I plan to provide a review worksheet and time next Tuesday to work on it. I am also happy to take questions about any previous homework problems.

Today: Examples with the Birch and Swinnerton-Dyer conjecture.

Summary of last time: If $E/\mathbb{Q}$ is an elliptic curve, it has an L -function $L(E, s) = \sum_{n=1}^{\infty} \frac{a_n(E)}{n^s}$.

The weak BSD conjecture predicts that if $E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \times \mathbb{Z}^r$, then $L(E, s)$ has a zero at $s = 1$ of order exactly r .

There is a number $\epsilon(E) \in \{\pm 1\}$ and a “completed L -function”

There is a number $\epsilon(E)$ so that $\Lambda(E, s) = N(E)^{s/2} \Gamma(s) L(E, s)$ so that $\Lambda(E, 2-s) = \epsilon(E) \Lambda(E, s)$. The order of the zero of $L(E, s)$ at $s = 1$ is odd if $\epsilon(E) = -1$ and even if $\epsilon(E) = 1$.

Theorem. *If $L(E, 1) \neq 0$, then the rank of $E(\mathbb{Q})$ is zero. If $L(E, 1) = 0$ but $L'(E, 1) \neq 0$ then the rank of $E(\mathbb{Q})$ is one.*

Strong BSD: Suppose that $E/\mathbb{Q}$ is an elliptic curve of rank r . Then

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\Omega \cdot \prod_p c_p(E) \cdot R(E/\mathbb{Q}) \cdot \#\text{III}(E/\mathbb{Q})}{(\#E(\mathbb{Q})_{\text{tors}})^2}.$$

Here Ω is either the real period (if the discriminant of $E < 0$) or twice the real period (if the discriminant of $E > 0$).

The $c_p(E)$ are the Tamagawa numbers.

The quantity $R(E/\mathbb{Q})$ is the regulator. To define this, let $P_1, P_2, \dots, P_r$ be r points of infinite order that (together with torsion points) generate $E(\mathbb{Q})$. For $P, Q \in E(\mathbb{Q})$, define the height pairing.

$$\langle P, Q \rangle = \frac{1}{2} \hat{h}(P + Q) - \frac{1}{2} \hat{h}(P) - \frac{1}{2} \hat{h}(Q).$$

Now, $R(E/\mathbb{Q})$ is the determinant of the $r \times r$ matrix whose ij th entry is $\langle P_i, P_j \rangle$.

The $\text{III}(E/\mathbb{Q})$ is the Shafarevich-Tate group.

The denominator is the square of the order of the torsion subgroup.

Example I: Let $E : x^3 + 1$. We have $\epsilon(E) = 1$. (The command in Magma is `RootNumber(E);`) Then $L(E, 1) \approx 0.7011$. We can get this by typing

```
E := EllipticCurve([0,0,0,0,1]);
L := LSeries(E);
Evaluate(L,1);
```

Since $L(E, 1) \neq 0$, the theorem above states that $E(\mathbb{Q})$ is finite. In fact, $E(\mathbb{Q}) \cong \mathbb{Z}/6\mathbb{Z} = \langle (2, 3) \rangle$, so E has exactly six rational points on it.

Example II: Let $E : y^2 = x^3 + 11192x$. We have $\epsilon(E) = -1$. This makes it so that $L(E, s)$ has an odd order zero at $s = 1$. In particular, $L(E, 1) = 0$. We have $L'(E, 1) \approx 41.3889$.

We can get this by typing

```
E := EllipticCurve([0,0,0,11192,0]);
L := LSeries(E : Precision := 5);
Evaluate(L,1 : Derivative := 1);
```

The theorem above implies that E has rank one. Magma is able to determine that $E(\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$ and is generated by $T = (0, 0)$ and a point P whose x -coordinate is

$$\left(\frac{6248894815165424517530051}{663852669320825316603460} \right)^2.$$

In the strong BSD conjecture, we have

$$\begin{aligned} \Omega &\approx 0.36052077 \\ c_2(E) &= 2 \\ c_{1399}(E) &= 2 \\ R(E/\mathbb{Q}) = \hat{h}(P) &\approx 114.80309... \\ (\#E(\mathbb{Q})_{\text{tors}})^2 &= 4. \end{aligned}$$

This would match with $\#\text{III}(E/\mathbb{Q}) = 1$, but computing $\text{III}(E/\mathbb{Q})$ is quite hard.

Example III: Let $E : y^2 = x^3 + 11197x$. Again we have $\epsilon(E) = -1$. If we run the command `rank(E);`

Magma returns 1 and true. The number 1 is a lower bound for the rank. The second return means that the rank is exactly 1.

```
G, mp := MordellWeilGroup(E);
```

The group returned is just $\mathbb{Z}/2\mathbb{Z}$. What's going on?

We can get more of an idea of what's going on if we run the command

```
MordellWeilShaInformation(E);
```

We can see that Magma knows the rank is 1 because the “analytic rank” is one (this is the order of vanishing of $L(E, s)$ at $s = 1$), but it can’t find a generator.

If we run

```
MordellWeilShaInformation(E : Effort := 2);
```

then it tries harder and finds a point on the curve with canonical height 313.667.

(If the conductor of the curve is higher, Magma won’t attempt to compute the analytic rank, and in a situation like this, `Rank(E)` would return 0 and false.)

Example IV: Let $E : y^2 = x^3 + 17$. Then $\epsilon(E) = 1$. We have $|L(E, 1)| < 10^{-30}$. This doesn’t prove that $L(E, 1) = 0$, but some theory allows one to verify that $L(E, 1) = 0$. The fact that $\epsilon(E) = 1$ means that $L(E, s)$ has an even order of vanishing at $s = 1$, and so $L'(E, 1) = 0$. Finally, $L''(E, 1) \approx 11.035381$. Thus, BSD predicts that E has rank 2. In fact,

$$E(\mathbb{Q}) \cong \mathbb{Z} \times \mathbb{Z} = \langle (-1, 4), (-2, 3) \rangle.$$

Example V: Let $E : y^2 + xy = x^3 - x^2 - 79x + 289$. This curve has $N(E) = 234446$. We have $\epsilon(E) = 1$. Moreover, $L(E, 1) = 0$ and so $L'(E, 1) = 0$. Numerically, we have $|L''(E, 1)| < 10^{-30}$ and $|L'''(E, 1)| < 10^{-30}$. We have $L^{(4)}(E, 1) \approx 214.65$. This suggests that $L(E, s)$ has a quadruple zero at $s = 1$, and so E should have rank 4.

In fact, E does have rank 4, and $E(\mathbb{Q})$ is generated by $(52/9, -95/27)$, $(55/9, -151/27)$, $(6, -5)$, and $(0, 17)$.

We don’t have an algorithm for determining whether $L^{(k)}(E, 1) = 0$ if $k \neq 0$ or 1. Hence, in some cases, we can’t actually calculate the order of the zero at $s = 1$ of $L(E, s)$.

Problem: Are there integers M , N and e , not all zero, for which

$$N^2 = 4133M^4 + 16532e^4?$$

If we set $x = M/e$ and $y = N/e^2$ we get $y^2 = 4133x^4 + 16532$. This hyperelliptic curve is locally solvable everywhere and point searching up to a bound of one million doesn’t find points.

This problem arises when we are computing the rank of $C : y^2 = x^2 - 4133^2x$. Here $C' : y^2 = x^3 + 4 \cdot 4133^2x$. There is a solution to the equation above if and only if 4133 is in the image of $\bar{\alpha}$. Magma tells us that 4133 is prime.

If we ask Magma for the rank of C , it tells us that the rank is 1 and the second return value is true. This means that $r = 1$ and so

$$2^1 = \frac{|\text{im } \alpha| |\text{im } \bar{\alpha}|}{4}$$

and so $|\text{im } \alpha| |\text{im } \bar{\alpha}| = 8$.

The image of α is contained in $\{\pm 1, \pm 4133\}$. The curve C has three points of order 2: $(-4133, 0)$, $(0, 0)$ and $(4133, 0)$ and the images of these under α are -4133 , -1 and $4133 \bmod \text{squares}$. So $|\text{im } \alpha| = 4$.

The image of $\bar{\alpha}$ is contained in $\{\pm 1, \pm 2, \pm 4133, \pm 8266\}$ but the negative values cannot occur since $y^2 = dx^4 + \frac{4 \cdot 4133^2}{d}$ has no solutions if $d < 0$.

We have $\bar{\alpha}(\bar{\mathcal{O}}) = 1$ and $\bar{\alpha}(\bar{T}) = 4 \cdot 4133^2 \equiv 1 \pmod{((\mathbb{Q}^\times)^2)}$.

We know that $|\text{im } \bar{\alpha}| = 2$ because the rank of C is 1.

If we test 2, we find that $y^2 = 2x^4 + 2 \cdot 4133^2$ has no points mod powers of 2 and so 2 is not in the image of $\bar{\alpha}$.

Likewise if we test 8266 we find that $y^2 = 8266x^4 + 8266$ has no points mod powers of 2 also.

Thus $\text{im } \bar{\alpha} = \{1, 4133\}$ and so there are solutions to the equation. We haven't found any though.

If we run the commands

```
C := EllipticCurve([0,0,0,-4133^2,0]);
ConjecturalRegulator(C);
```

Magma tells us that for our elliptic curve C , assuming strong BSD,

$$R(E/\mathbb{Q}) \cdot \#\text{III}(E/\mathbb{Q}) = 771.811435.$$

It turns out that $C(\mathbb{Q})$ has rank 1 and a generator P has $\hat{h}(P) \approx 771.81$. Remember that $\hat{h}(P) \approx h(P)$ and so the generator is a point $P \in C(\mathbb{Q})$ for which the *log* of the numerator and denominator are about 771.81. So the numerator and denominator should have size about $e^{771.81}$.

My notes from 2017 say that if you run `MordellWeilShaInformation` with `Effort := 2`, that Magma will find a generator after about 16 hours.